

CUSTOMER SHOPPING BEHAVIOR ANALYTICS

End-to-End Data Analyst Portfolio Project

Python (Pandas) • MySQL • Power BI

Project Report

Project element	Details
Dataset	Customer Shopping Behavior (3,900 records)
Prepared dataset	Cleaned dataset with 3,900 records and 19 fields
Data preparation	Python / Pandas in VS Code
Database analysis	MySQL
Visualization	Microsoft Power BI
Primary objective	Analyze customer purchasing behavior and derive actionable business insights

1. Executive Summary

This project analyzes customer shopping behavior using an end-to-end analytics workflow. The raw CSV data was first cleaned and transformed with Python/Pandas, then the prepared dataset was loaded into MySQL for business-oriented SQL analysis. The analyzed data was subsequently connected to Power BI to create an interactive customer behavior dashboard.

The dataset contains 3,900 customer-level purchase records and 3,900 unique customer IDs. Total recorded purchase revenue is \$233,081, with an average purchase amount of \$59.76 and an average review rating of 3.75.

The analysis focuses on gender revenue, discount behavior, product ratings, shipping type, subscription behavior, discount penetration, customer segmentation, product rankings within categories, repeat buyers, and age-group revenue.

2. Business Objective

- Understand customer purchasing patterns and revenue contribution.
- Identify high-performing products and categories.
- Evaluate whether subscribed customers spend more than non-subscribers.
- Compare purchasing behavior across shipping types and age groups.
- Understand discount usage and products with high discount penetration.
- Segment customers using previous purchase behavior.
- Provide business insights that can support customer engagement, promotions, and product strategy.

3. Tools & Technology Stack

Tool	Purpose
Python / Pandas	Data cleaning, transformation, feature engineering and preparation
VS Code	Python development environment
MySQL	Data storage and SQL-based business analysis
MySQL Workbench	Database management and query execution
Power BI	Interactive dashboarding and visualization
CSV	Raw and cleaned data exchange format

4. Data Preparation & Transformation

The original dataset contained 3,900 records and 18 columns. The Pandas workflow transformed it into a 19-column cleaned dataset without changing the number of records.

Transformation	Implementation / Result
Missing review ratings	Filled missing Review Rating values using the median review rating within each Category.
Column naming	Converted column names to lowercase and replaced spaces with underscores.
Column standardization	Renamed purchase_amount_(usd) to purchase_amount.
Age grouping	Created age_group using quartile-based qcut with Young Adult, Adult, Middle Age and Senior labels.
Purchase frequency	Mapped frequency labels to numeric purchase_frequency_days (7, 14, 30, 90, 365).
Redundant field removal	Verified discount_applied and promo_code_used were identical for all rows, then removed promo_code_used.

Output	Saved the transformed dataset as clean_data.csv.
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The cleaning script explicitly performs these transformations, including category-level median imputation, column standardization, age-group creation, purchase-frequency mapping, and removal of the redundant promo-code field.

5. Data Overview

Metric	Value
Rows	3,900
Columns after cleaning	19
Unique customer IDs	3,900
Total purchase amount	\$233,081
Average purchase amount	\$59.76
Average review rating	3.75
Discount applied	43% of records
Subscription status = Yes	27% of records

6. SQL Business Analysis

The MySQL analysis contains 10 business questions. The queries use aggregation, conditional logic, subqueries, CTEs, and window functions to answer operational and customer-behavior questions.

#	Business question	SQL technique
1	Total revenue generated by male vs female customers	GROUP BY + SUM
2	Customers receiving discounts who purchased above average	WHERE + subquery
3	Top five products by average review rating	GROUP BY + AVG + ORDER BY + LIMIT
4	Average purchase amount for Standard vs Express shipping	WHERE + GROUP BY + AVG
5	Subscriber vs non-subscriber customer count, spend and revenue	GROUP BY + COUNT + AVG + SUM
6	Top five products by percentage of purchases with discount	CASE WHEN + COUNT
7	New, Returning and Loyal customer segmentation	CTE + CASE
8	Top three most-purchased products within each category	CTE + ROW_NUMBER window function
9	Repeat buyers with more than five previous purchases by subscription status	WHERE + GROUP BY
10	Revenue contribution by age group	GROUP BY + SUM

7. Key Findings

7.1 Revenue by Gender

- Male: \$157,890
- Female: \$75,191

Male customers generated \$157,890, compared with \$75,191 from female customers. The difference is driven by the composition of the records in this dataset and should not automatically be interpreted as a causal preference.

7.2 Subscription Behavior

- No: 2,847 customers, average spend \$59.87, revenue \$170,436.
- Yes: 1,053 customers, average spend \$59.49, revenue \$62,645.

In this dataset, non-subscribers generate more total revenue because they represent a much larger customer count. Average spend is very similar between the two groups, so subscription status alone does not show a meaningful average-spend advantage.

7.3 Shipping Type

- Express: average purchase \$60.48
- Standard: average purchase \$58.46

Express shipping has a higher average purchase amount (\$60.48) than Standard shipping (\$58.46) in the analyzed records.

7.4 Top-Rated Products

Product	Average rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Handbag	3.78

7.5 Products with Highest Discount Penetration

Product	Discounted purchase %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

7.6 Customer Segmentation

Segment	Customers
Loyal	3,116
Returning	701
New	83

The project classifies customers using previous purchases: 1 purchase as New, 2–10 as Returning, and more than 10 as Loyal. The resulting distribution is heavily weighted toward the Loyal segment.

7.7 Repeat Buyers and Subscription

- No: 2,518 repeat buyers with more than 5 previous purchases
- Yes: 958 repeat buyers with more than 5 previous purchases

7.8 Revenue by Age Group

Age group	Revenue
Young Adult	\$62,143
Middle Age	\$59,197
Adult	\$55,978
Senior	\$55,763

8. Power BI Dashboard

The cleaned/analyzed data was loaded into Power BI and used to build the Customer Behavior Dashboard. The dashboard is the final visualization layer of the project and is intended to make the SQL findings easier for a business stakeholder to explore interactively.

- Use KPI cards to communicate headline measures such as customer count, revenue, average purchase amount and ratings.
- Use category/product visuals to identify high-performing items.
- Use customer and subscription visuals to compare behavioral groups.
- Use demographic and geographic views to explore customer composition.
- Use filters/slicers to allow stakeholders to drill into customer, product, season, category and subscription segments.

Note: the supplied PBIX file was included as the dashboard artifact; this report does not infer visual names or measures that could not be directly verified from the PBIX binary.

9. Business Recommendations

1. Investigate the large customer-base difference between subscribers and non-subscribers and test targeted subscription incentives rather than assuming subscribers inherently spend more.
2. Review products with high discount penetration to determine whether discounts are driving incremental demand or simply reducing realized revenue.
3. Use the highest-rated products in merchandising and promotional campaigns, while monitoring whether high ratings are accompanied by strong purchase volume.
4. Monitor Express-shipping orders because their average purchase amount is higher in this dataset; evaluate whether premium shipping is associated with larger baskets.
5. Prioritize retention and engagement of Returning customers to move them into the Loyal segment.
6. Use age-group revenue differences to tailor promotions and product messaging, while validating findings with larger or time-based datasets before making strategic decisions.

10. Project Workflow

Raw CSV → Python/Pandas cleaning & feature engineering → clean_data.csv → MySQL import → SQL business analysis → Power BI dashboard → insights & recommendations

11. Limitations & Future Improvements

- The dataset is a static snapshot, so time-series trends such as monthly revenue growth cannot be evaluated from the available fields.
- The dataset contains purchase amount but no explicit cost/profit field, so profitability and margin analysis are not possible.
- Customer IDs are unique in the supplied data, limiting true customer-level longitudinal analysis despite the previous_purchases field.
- Future versions could add order dates, product cost, quantity, returns/refunds, acquisition channel, and customer cohorts.
- Future analysis could include RFM segmentation, cohort retention, customer lifetime value, and profitability analysis.

12. Portfolio / Resume Description

Suggested project title:

Customer Shopping Behavior Analytics | Python • SQL • Power BI

Developed an end-to-end customer analytics project by cleaning and transforming 3,900 shopping records using Python/Pandas, loading the prepared dataset into MySQL, answering 10 business questions using SQL aggregations, subqueries, CTEs and window functions, and building an interactive Power BI dashboard to communicate customer, product, subscription and purchasing insights.

Suggested resume bullets:

- Cleaned and transformed 3,900 customer purchase records using Pandas, including category-level median imputation, column standardization, feature engineering and redundant-field removal.
- Loaded the prepared dataset into MySQL and performed 10 business analyses using GROUP BY, CASE, subqueries, CTEs and ROW_NUMBER() window functions.
- Built a Power BI customer behavior dashboard to analyze revenue, customer segments, product performance, discounts, shipping and subscription behavior.

Appendix A — Final Dataset Fields

customer_id, age, gender, item_purchased, category, purchase_amount, location, size, color, season, review_rating, subscription_status, shipping_type, discount_applied, previous_purchases, payment_method, frequency_of_purchases, age_group, purchase_frequency_days

Appendix B — Source Artifacts

- customer_shopping_behavior.csv — original dataset
- Data_Cleaning.py — Pandas data preparation script
- clean_data.csv — transformed dataset
- Analysed_data.sql — MySQL database and business analysis queries
- Customer Behavior Dashboard.pbix — Power BI dashboard artifact